

Deep Reinforcement Learning-based Dynamic TWT Scheduling for Heterogeneous Wi-Fi Networks

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The 35th International Conference on
Computer Communications and Networks (ICCCN'26)



Outline

- Motivation
- Target Wake Time (TWT) in IEEE 802.11ax
- System Model
- State, Action, Reward
- Simulation Platform
- Model
- Train and Evaluation results

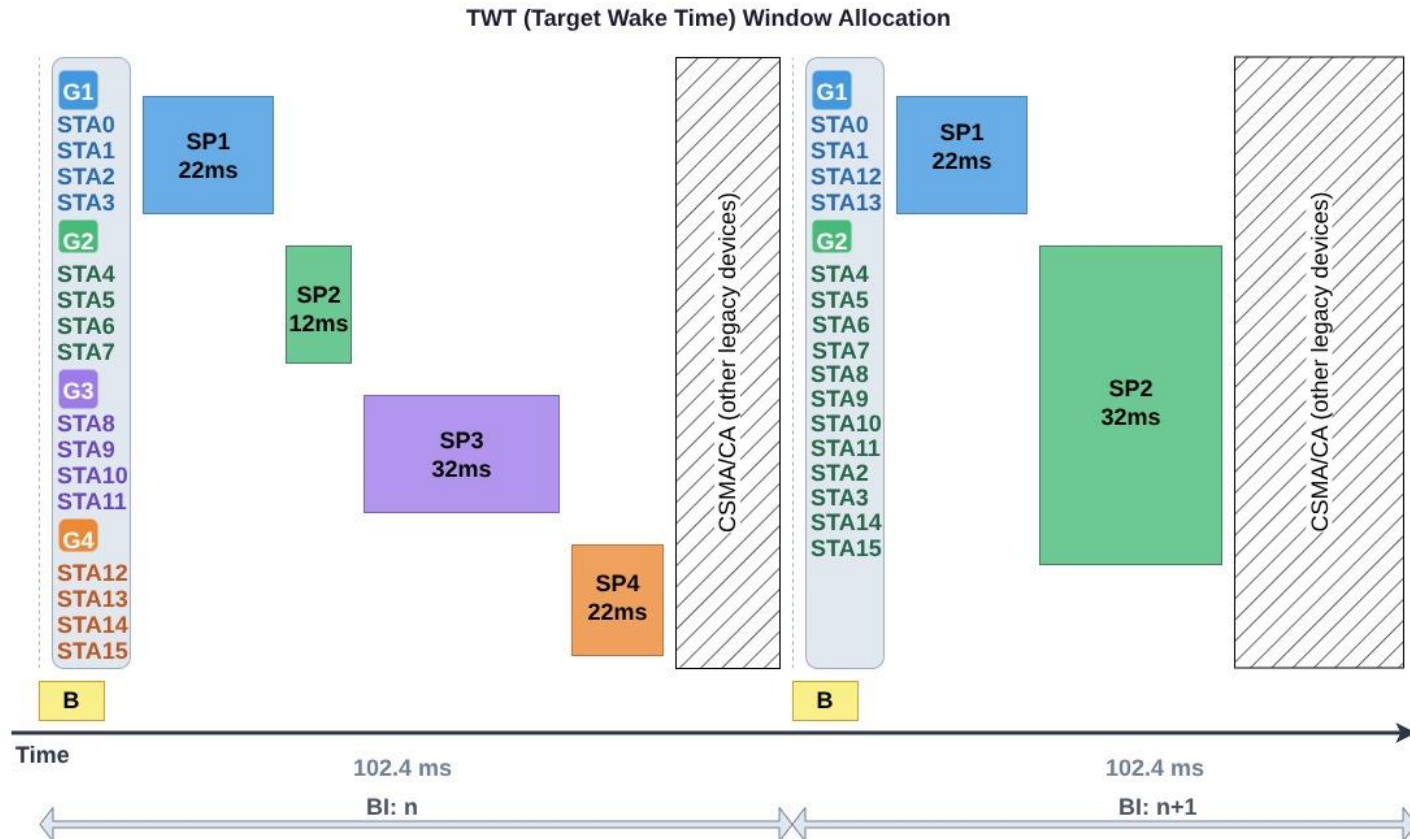
Motivation: Future Heterogeneous Wi-Fi Needs Smart Medium Access

- Different QoS simultaneously
- AP must serve all of them
- Legacy CSMA/CA offers no coordination across classes

Class	Rate	Latency	Battery
IoT sensor	256 kbps	5 s	yes
Video camera	2 Mbps	500 ms	no
Voice assistant	64 kbps	50 ms	no
Video stream	5 Mbps	100 ms	no

- Wasted energy on *contention, idle listening* (50mA / 0.12mA for idle/sleep)
- **Industry Response: Target Wake Time (TWT) in Wi-Fi 6**

Primer on Target Wake Time (TWT)



- AP partitions N STAs into $K \leq K_{max}$ groups
- Each group is assigned a Service Period (SP): duration + offset from Beacon
- Must satisfy $\sum_{k=1}^K d_k + (K-1) \delta_g \leq T_{BI} - T_{res}$, $N = 16, K_{max} = 4, \delta_g = 2ms, T_{res} = 7.4ms$

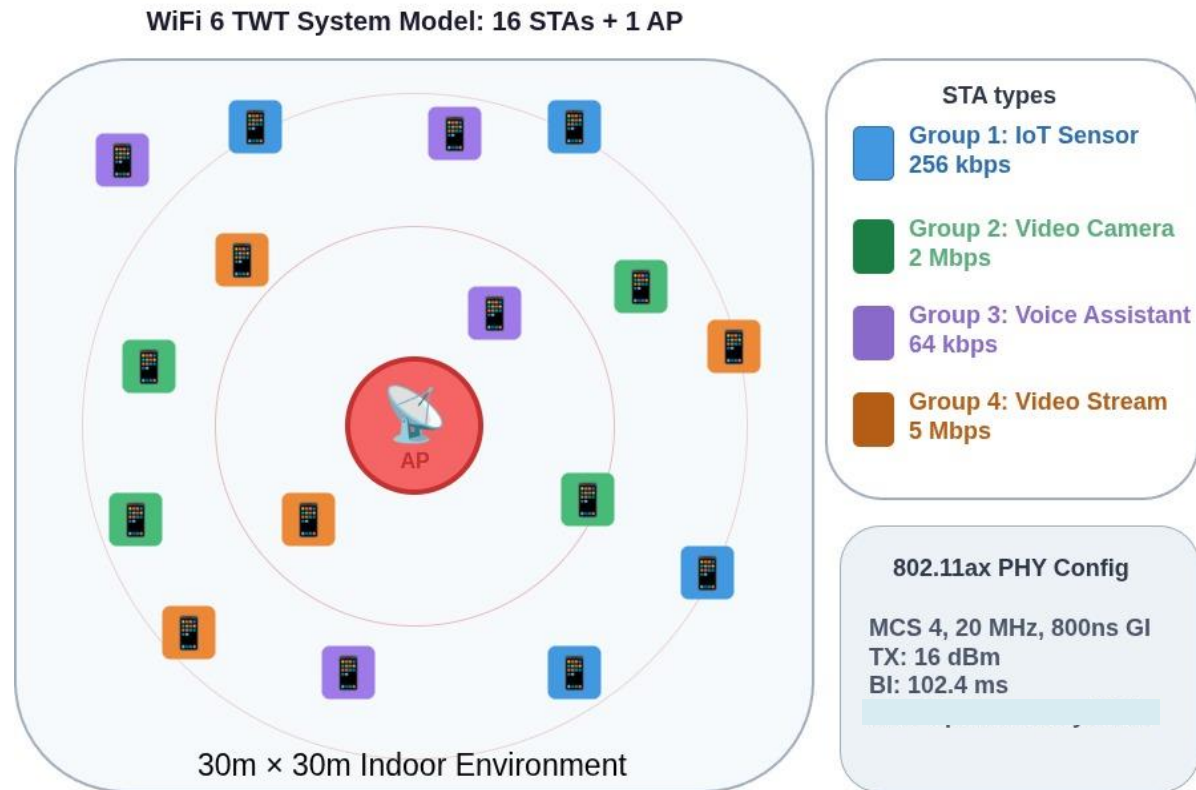
Challenge: TWT Scheduling is NP Hard

- With **16 STAs** and **up to 4 groups**, STA-to-group assignment alone is $4^{16} \approx 4.3$ billion
 - Per-group wake durations on top of that pushes the joint space past $\sim 10^{14}$
 - No closed form for optimal solution
 - Heuristics/Analytical do not respond well to dynamic traffic + no optimality guarantee
- Propose: RL-driven TWT scheduler
 - Also: (No existing work offers ALL)
 - Protocol Compliant
 - End-to-end full fidelity ns-3 simulation
 - Dynamic + Heterogenous network

Question: Can the AP learn, from protocol-compliant metrics, under dynamic heterogenous traffic, TWT schedules (grouping + assignment) to balance throughput, energy, and latency?

System Model

- Unilateral TWT assignment from AP
- 5 GHz band
- $N = 16$ STAs
- $C = 4$ device groups
- $\alpha=3$ + Nakagami ($m_0=3.0$, $m_1=1.5$)
- Energy at 3 V: TX 232 / RX 66 / idle 50 / sleep 0.12 mA
- TWT update/20 BI
- 35 updates/Episode
- AP is blind to STA classes



Observation Space: Two types

PER-STA METRICS EXPORTED AT EACH DECISION STEP.

Realistic (Obs + Reward)	Oracle (Reward Only)
<i>Protocol-observed (5):</i> bsr_queue_index, airtime_used, fcs_errors, rx_fragments, last_rx_timestamp	<i>Instantaneous (2):</i> queue_size_bytes, queue_size_packets
<i>AP-derived (4):</i> awake_time, sleep_time, duty_cycle, packets_tx	<i>Deltas (4):</i> Δ bytes_tx, Δ energy, Δ mpdu_drops, Δ pkts_enqueued
<i>Computed deltas (4):</i> Δ airtime, Δ awake, Δ sleep, Δ packets_tx	
13 feat $\times$ N STAs	6 feat $\times$ N STAs

- Realistic + Oracle metrics observed by reward signal during training only
- Realistic metrics observed by the RL agent after deployment

Action Space: ~1E14 to 380

SCHEDULE DIMENSION ($|\mathcal{A}_s|=20$): GROUP-TIMING CONFIGURATIONS

ID	Type	K	Wake durations (ms)
0–4	Single group	1	$d \in \{90, 80, 60, 40, 20\}$
5–9	Equal 2-group	2	$d_k \in \{45, 40, 30, 20, 15\}$
10–13	Unequal 2-group	2	60+20, 50+30, 40+20, 30+15
14–16	3-group	3	30+30+25, 40+25+20, 25×3
17–19	4-group	4	22×4, 30+20+20+15, 20+20+15+15

TABLE IV

ASSIGNMENT DIMENSION ($|\mathcal{A}_m|=19$): STA-TO-GROUP MAPPINGS.

ID	Type	Pattern	K_{req}
0–3	All-to-one	All STAs $\rightarrow$ group k	1
4–6	Round-robin	Cyclic, $k \in \{2, 3, 4\}$ groups	2–4
7–8	Half-split	Sequential / even-odd	2
9	Thirds-split	Sequential 3-way	3
10	Modulo-3	$i \bmod 3$	3
11	Quarters-split	Sequential 4-way	4
12	Modulo-4	$i \bmod 4$	4
13–14	Weighted (2g)	75/25, 60/40	2
15	Weighted (3g)	50/30/20	3
16	Weighted (4g)	40/30/20/10	4
17–18	Interleave	Step-2, step-4	2

- 20 Schedules X 19 Assignments
- Multi-Discrete action
- 2 independent categorical heads
- 380 possible discrete actions
- Discrete but designed to *span* the whole space

Reward Function: 6 components, 3 presets

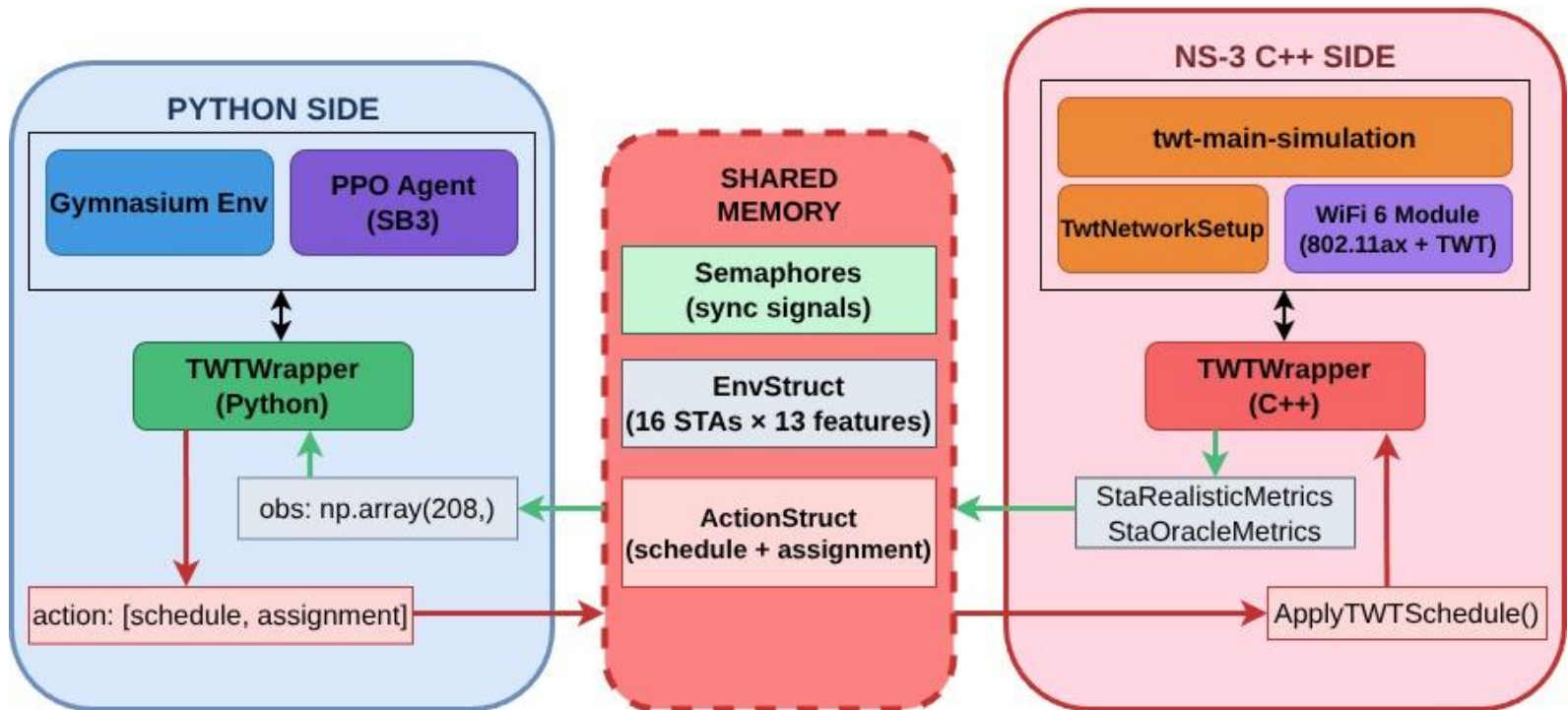
- R_c bounded via $\tanh$
- R_c consists of up to 3 elements:
 - Z-scored aggregates
 - Fairness term
 - Count based penalties
- Z-score + other params obtained thru 50,000-step EDA under uniform random policy
- Weights for each preset and component are:

$$r_t = \sum_{c \in C} w_c \bullet R_c$$

Preset	Thruput	Queue	Energy	Drops	Air	Channel
Throughput	0.35	0.20	0.10	0.15	0.15	0.05
Energy	0.20	0.10	0.35	0.10	0.20	0.05
Queue	0.20	0.35	0.10	0.20	0.10	0.05

Simulation Platform: NS3ai+Python integration

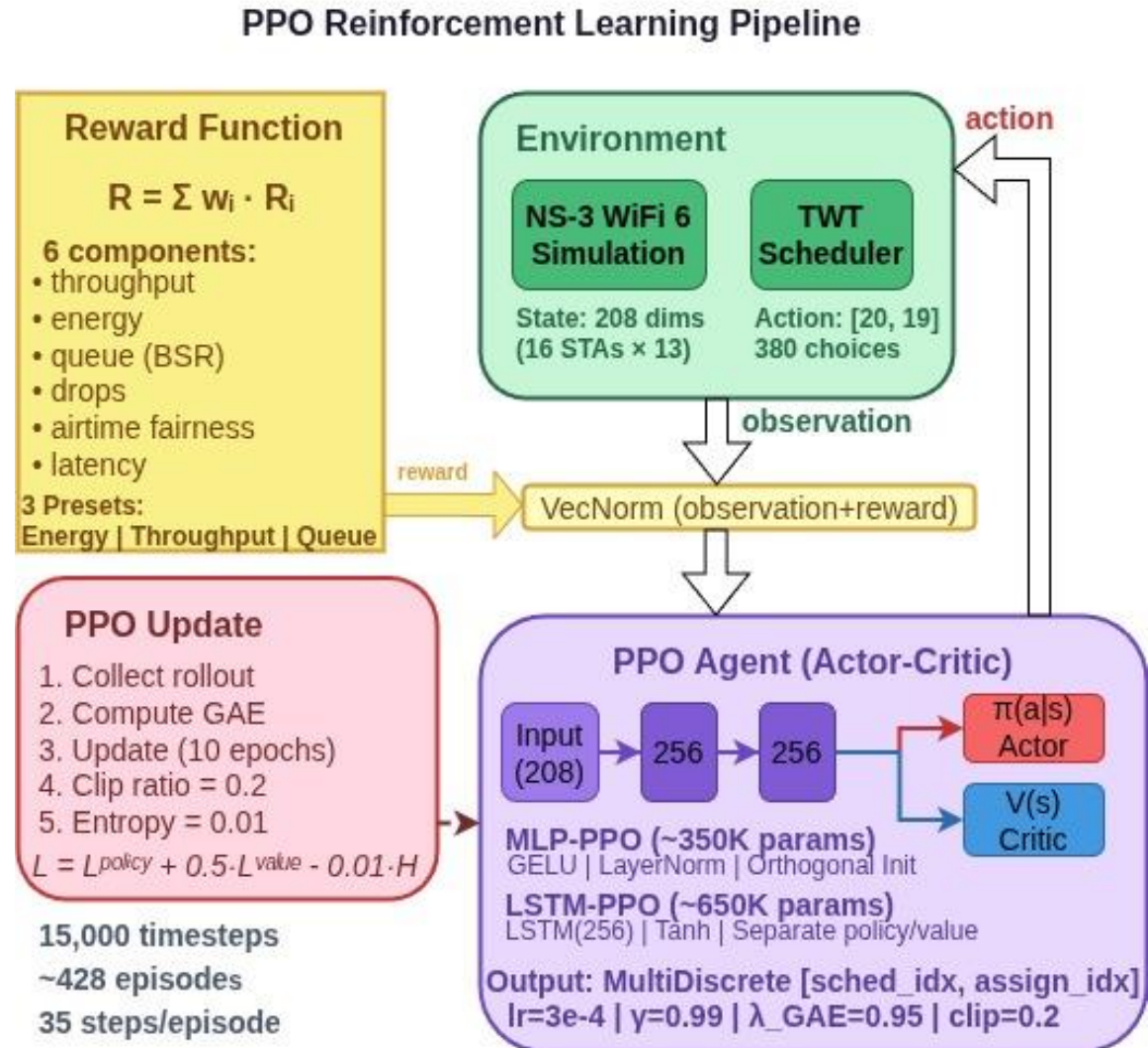
Python ↔ NS-3 Communication Bridge



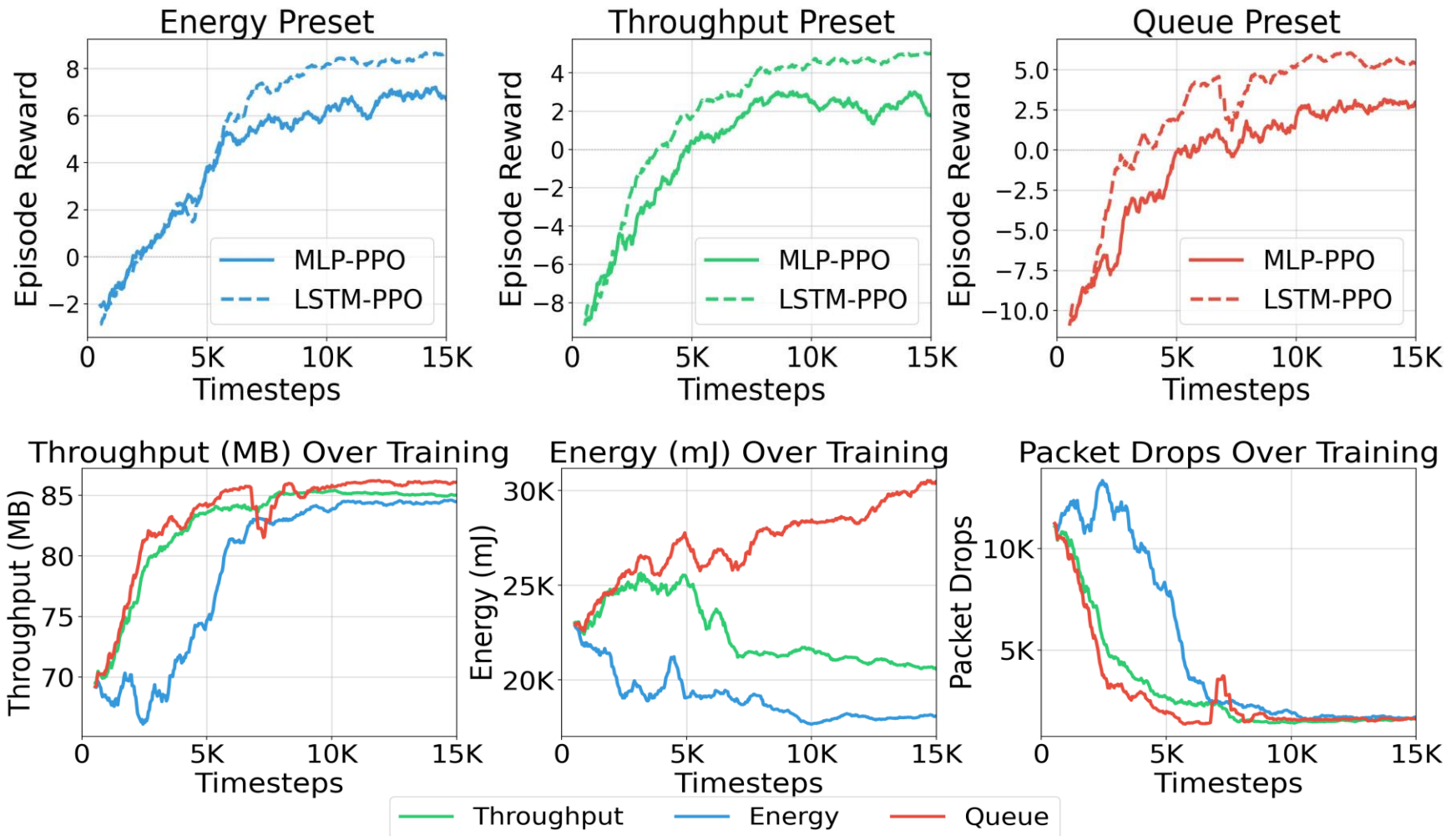
- ns-3: Full fidelity 802.11ax MAC/PHY || Python: PPO (SB3)
- 1 Episode = 35 steps (per 20 BIs); 850 beacon intervals (≈ 87.6 s)
- Dynamic: topology re-randomized between episodes (device class, position, traffic config, seeds)

RL model: MLP-PPO / LSTM-PPO

- Running VecNorm on top of EDA norm
- Weights are orthogonally initialized for improved gradient flow
- 2 arch, same PPO: memoryless MLP vs recurrent LSTM
- LSTM keeps a hidden state across 35 steps to track trends



Training



- Energy preset – best reward
- LSTM consistently better during training

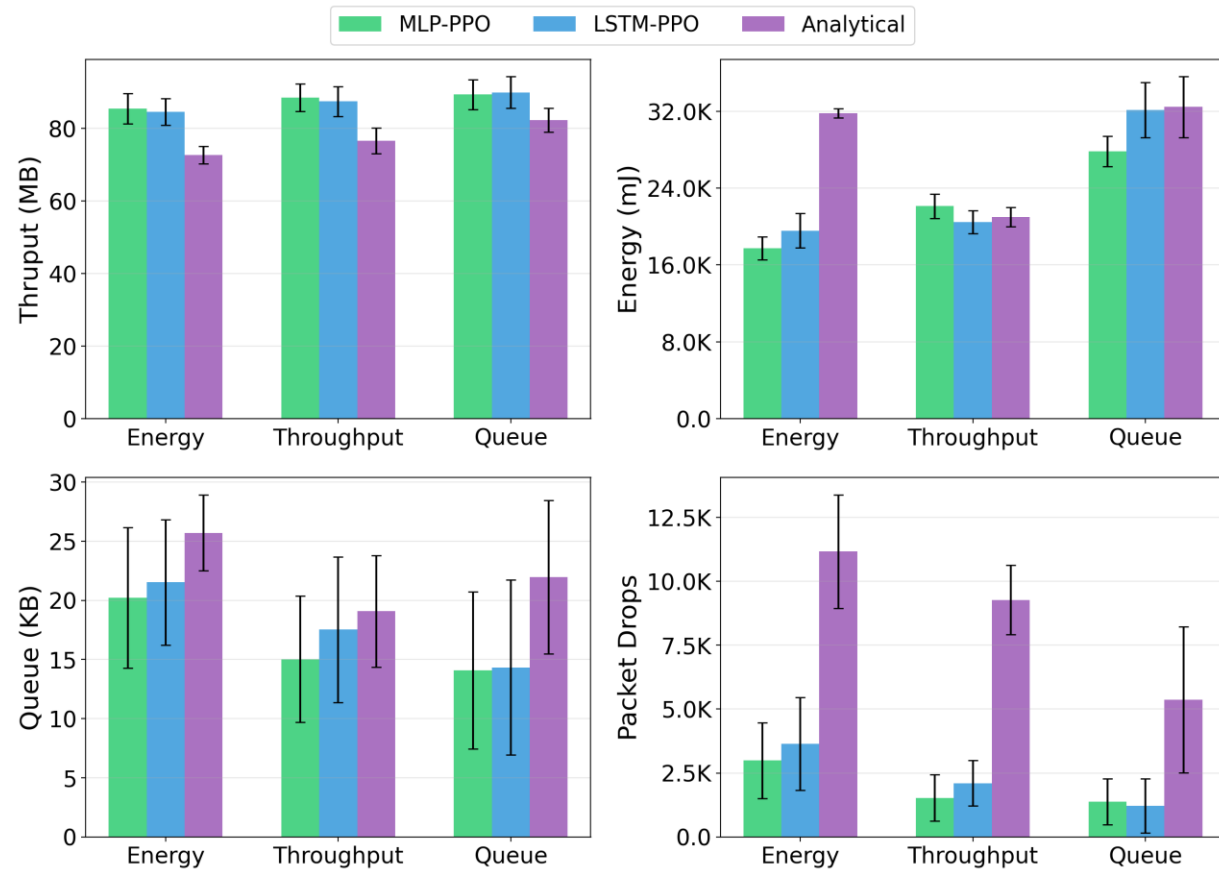
Evaluation

- {MLP, LSTM, Analytical} X {3 presets} – 50 episodes each
- Both RL models beat baseline
- MLP-PPO vs Baseline: (energy)

–44% energy

+18% throughput

–73% drops



Analytical Baseline

- M/D/1 queue + energy model
- Calibrated from same EDA

	Reward	MB	mj	Drops
MLP-PPO	7.50	85.5	17,745	2,987
LSTM-PPO	5.84	84.6	19,575	3,644
Analytical	-14.3	72.7	31,789	11,170

Summary of energy preset Eval on 50 episodes

Conclusion

- Adaptive, RL driven TWT scheduling for heterogeneous dynamic network
- Protocol compliant observation, full fidelity simulation; deployable
- Both MLP and LSTM PPO beat M/D/1 baseline
- Future Work:
 - Expand action space
 - More capable model+architecture
 - Scale up number of devices and nlasses
 - Joint TWT-OFDMA-MLO optimization

Thank you!

Questions?



<https://github.com/TXST-SHINe-Lab/802.11ax-DRL-TWT-ICCCN26>

